Project Proposal: Stock Market Prediction System Using Reinforcement Learning

Minjae Kyung

Current State

The stock market is unpredictable; there are too many variables and too many unforeseeable situations. However, this led to the thought, "Couldn't a machine learning model predict it?" So, I am trying to build a model using RL.

Model Development

In this project, we aim to create an algorithm that predicts the stock chart for each individual stock. The goal of this model is to first collect various news articles from the previous day, embed them, and calculate the average of the resulting vectors. This average vector is then fed into the model along with the stock data to predict the stock price for the next day.

Data Collection

To do this, we will retrieve stock data using an API from Yahoo! Finance (https://finance.yahoo.com/) and collect news information using NewsAPI (https://newsapi.org/) or the GDELT Project (https://www.gdeltproject.org/). For this, we plan to use Python modules like yfinance and gdeltdoc.

Evaluation and Validation

To compare the change predicted by the model with the actual change, we will use metrics such as RMSE and R2 to measure the difference between the two. We also plan to compare the rate of return from investing based on the model with the rate of return from a simple buy-and-hold strategy.

Reference

Awad, A. L., Elkaffas, S. M., & Fakhr, M. W. (2023). Stock Market Prediction Using Deep Reinforcement Learning. *Applied System Innovation*. https://doi.org/10.3390/asi6060106